

Лабораторная работа

Номер 8

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Приобретение практических навыков по установке и конфигурированию SMTP-сервера

```
[nsandryushin@server.nsandryushin.net ~]$ sudo -i
[sudo] password for nsandryushin:
[root@server.nsandryushin.net ~]# dnf -y install postfix
Extra Packages for Enterprise Linux 10 - x86_64      65 kB/s | 36 kB    00:00
Extra Packages for Enterprise Linux 10 - x86_64      2.8 MB/s | 4.8 MB  00:01
Rocky Linux 10 - BaseOS                             8.7 kB/s | 4.3 kB  00:00
Rocky Linux 10 - BaseOS                             3.5 MB/s | 21 MB   00:06
Rocky Linux 10 - AppStream                          14 kB/s | 4.3 kB   00:00
Rocky Linux 10 - AppStream                         476 kB/s | 2.2 MB  00:04
Rocky Linux 10 - CRB                                11 kB/s | 4.3 kB   00:00
Rocky Linux 10 - CRB                               1.2 MB/s | 527 kB  00:00
Rocky Linux 10 - Extras                             9.1 kB/s | 3.1 kB   00:00
Rocky Linux 10 - Extras                             12 kB/s | 5.4 kB   00:00
Dependencies resolved.
=====
Package            Architecture Version            Repository        Size
=====
Installing:
postfix            x86_64        2:3.8.5-8.el10    appstream         1.5 M
Installing dependencies:
postfix-lmdb       x86_64        2:3.8.5-8.el10    appstream          25 k
=====
Transaction Summary
=====
Install 2 Packages

Total download size: 1.5 M
Installed size: 4.5 M
Downloading Packages:
(1/2): postfix-lmdb-3.8.5-8.el10.x86_64.rpm        563 kB/s | 25 kB    00:00
(2/2): postfix-3.8.5-8.el10.x86_64.rpm             6.0 MB/s | 1.5 MB   00:00
=====
Total                                           2.7 MB/s | 1.5 MB   00:00
```

Рис. 1: Установка Postfix

Установка s-nail

```
[root@server.nsandryushin.net ~]# dnf -y install s-nail
Last metadata expiration check: 0:00:40 ago on Sat 25 Oct 2025 02:26:38 PM UTC.
Dependencies resolved.
=====
Package           Architecture      Version            Repository          Size
=====
Installing:
s-nail             x86_64            14.9.24-12.el10    appstream           633 k
=====

Transaction Summary
=====
Install 1 Package

Total download size: 633 k
Installed size: 1.2 M
Downloading Packages:
s-nail-14.9.24-12.el10.x86_64.rpm                2.0 MB/s | 633 kB    00:00
-----
Total                                              859 kB/s | 633 kB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                :                               1/1
  Running scriptlet: s-nail-14.9.24-12.el10.x86_64         1/1
  Installing           : s-nail-14.9.24-12.el10.x86_64         1/1
  Running scriptlet: s-nail-14.9.24-12.el10.x86_64         1/1
```

Рис. 2: Установка s-nail

Предварительная настройка

```
[root@server.nsandryushin.net ~]# firewall-cmd --add-service=smtp
success
[root@server.nsandryushin.net ~]# firewall-cmd --add-service=smtp --permanent
success
[root@server.nsandryushin.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https smtp ssh ssh-custom
[root@server.nsandryushin.net ~]# restorecon -vR /etc
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0
[root@server.nsandryushin.net ~]# systemctl enable postfix
Created symlink '/etc/systemd/system/multi-user.target.wants/postfix.service' → '/usr/lib/systemd/system/postfix.service'.
[root@server.nsandryushin.net ~]# systemctl start postfix
```

Рис. 3: Предварительная настройка

```
[root@server.nsandryushin.net ~]# postconf
2bounce_notice_recipient = postmaster
access_map_defer_code = 450
access_map_reject_code = 554
address_verify_cache_cleanup_interval = 12h
address_verify_default_transport = $default_transport
address_verify_local_transport = $local_transport
address_verify_map = btree:$data_directory/verify_cache
address_verify_negative_cache = yes
address_verify_negative_expire_time = 3d
address_verify_negative_refresh_time = 3h
address_verify_pending_request_limit = 5000
address_verify_poll_count = ${stress?{1}:{3}}
address_verify_poll_delay = 3s
address_verify_positive_expire_time = 31d
address_verify_positive_refresh_time = 7d
address_verify_relay_transport = $relay_transport
address_verify_relayhost = $relayhost
address_verify_sender = $double_bounce_sender
address_verify_sender_dependent_default_transport_maps = $sender_dependent_default_transpo
rt_maps
address_verify_sender_dependent_relayhost_maps = $sender_dependent_relayhost_maps
address_verify_sender_ttl = 0s
address_verify_service_name = verify
address_verify_transport_maps = $transport_maps
address_verify_virtual_transport = $virtual_transport
alias_database = lmbd:/etc/aliases
alias_maps = lmbd:/etc/aliases
allow_mail_to_commands = alias, forward
allow_mail_to_files = alias, forward
```

Рис. 4: postconf

Обновление конфигурации postfix

```
[root@server.nsandryushin.net ~]# postconf myorigin
myorigin = $myhostname
[root@server.nsandryushin.net ~]# postconf mydomain
mydomain = nsandryushin.net
[root@server.nsandryushin.net ~]# postconf -e 'myorigin = $mydomain'
postconf: fatal: missing '=' after attribute name: "???myorigin"
[root@server.nsandryushin.net ~]# postconf -e 'myorigin = $mydomain'
postconf: fatal: missing '=' after attribute name: "???myorigin"
[root@server.nsandryushin.net ~]# postconf -e 'myorigin = $mydomain'
[root@server.nsandryushin.net ~]# postconf myorigin
myorigin = $mydomain
[root@server.nsandryushin.net ~]# postfix check
[root@server.nsandryushin.net ~]# systemctl reload postfix
```

Рис. 5: Обновление конфигурации postfix

```
[root@server.nsandryushin.net ~]# postconf -n
alias_database = ldap:/etc/aliases
alias_maps = ldap:/etc/aliases
command_directory = /usr/sbin
compatibility_level = 3.8
daemon_directory = /usr/libexec/postfix
data_directory = /var/lib/postfix
debug_peer_level = 2
debugger_command = PATH=/bin:/usr/bin:/usr/local/bin:/usr/X11R6/bin ddd $daemon_directory/
$process_name $process_id & sleep 5
default_database_type = ldap
html_directory = no
inet_interfaces = localhost
inet_protocols = all
mail_owner = postfix
mailq_path = /usr/bin/mailq.postfix
manpage_directory = /usr/share/man
meta_directory = /etc/postfix
mydestination = $myhostname, localhost.$mydomain, localhost
myorigin = $mydomain
newaliases_path = /usr/bin/newaliases.postfix
queue_directory = /var/spool/postfix
readme_directory = /usr/share/doc/postfix/README_FILES
sample_directory = /usr/share/doc/postfix/samples
sendmail_path = /usr/sbin/sendmail.postfix
setgid_group = postdrop
shlib_directory = /usr/lib64/postfix
smtp_tls_CAfile = /etc/pki/tls/certs/ca-bundle.crt
smtp_tls_CAspath = /etc/pki/tls/certs
smtp_tls_security_level = may
```

Рис. 6: postfix -n

```
[root@server.nsandryushin.net ~]# postconf -e 'mydomain = nsandryushin.net'
[root@server.nsandryushin.net ~]# postconf inet_protocols
inet_protocols = all
[root@server.nsandryushin.net ~]# postconf -e 'inet_protocols = ipv4'
[root@server.nsandryushin.net ~]# postfix check
[root@server.nsandryushin.net ~]# systemctl reload postfix
```

Рис. 7: Обновление настроек протоколов

```
[nsandryushin@server.nsandryushin.net ~]$ echo . | mail -s test1 nsandryushin@server.nsandryushin.net  
[nsandryushin@server.nsandryushin.net ~]$ █
```

Рис. 8: Отправка письма

```
[nsandryushin@server.nsandryushin.net ~]$ sudo tail -f /var/log/maillog
[sudo] password for nsandryushin:
Oct 25 14:32:06 server postfix/postfix-script[10881]: refreshing the Postfix mail system
Oct 25 14:32:07 server postfix/master[10615]: reload -- version 3.8.5, configuration /etc/postfix
Oct 25 14:32:07 server postfix/master[10615]: warning: ignoring inet_protocols parameter value change
Oct 25 14:32:07 server postfix/master[10615]: warning: old value: "all", new value: "ipv4"
Oct 25 14:32:07 server postfix/master[10615]: warning: to change inet_protocols, stop and start Postfix
Oct 25 14:32:48 server postfix/pickup[10885]: 3AAC220444A4: uid=1001 from=<nsandryushin>
Oct 25 14:32:48 server postfix/cleanup[10930]: 3AAC220444A4: message-id=<20251025143248.3AAC220444A4@server.nsandryushin.net>
Oct 25 14:32:48 server postfix/qmgr[10886]: 3AAC220444A4: from=<nsandryushin@nsandryushin.net>, size=358, nrcpt=1 (queue active)
Oct 25 14:32:48 server postfix/local[10933]: 3AAC220444A4: to=<nsandryushin@server.nsandryushin.net>, relay=local, delay=0.12, delays=0.08/0.03/0/0.01, dsn=2.0.0, status=sent (delivered to mailbox)
Oct 25 14:32:48 server postfix/qmgr[10886]: 3AAC220444A4: removed
```

Рис. 9: Лог почты

```
[nsandryushin@server.nsandryushin.net ~]$ sudo ls /var/spool/mail/  
[sudo] password for nsandryushin:  
nsandryushin  vagrant  
[nsandryushin@server.nsandryushin.net ~]$ █
```

Рис. 10: /var/spool/mail

```
[nsandryushin@client.nsandryushin.net ~]$ sudo -i

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for nsandryushin:
[root@client.nsandryushin.net ~]# dnf -y install postfix
Extra Packages for Enterprise Linux 10 - x86_64          6.6 kB/s | 36 kB      00:05
Extra Packages for Enterprise Linux 10 - x86_64          1.6 MB/s | 4.8 MB     00:03
Rocky Linux 10 - BaseOS                                  10 kB/s | 4.3 kB      00:00
Rocky Linux 10 - BaseOS                                  204 kB/s | 21 MB      01:45
Rocky Linux 10 - AppStream                               9.0 kB/s | 4.3 kB      00:00
Rocky Linux 10 - AppStream                              139 kB/s | 2.2 MB      00:16
Rocky Linux 10 - CRB                                     12 kB/s | 4.3 kB      00:00
Rocky Linux 10 - CRB                                    687 kB/s | 527 kB     00:00
Rocky Linux 10 - Extras                                  8.7 kB/s | 3.1 kB      00:00
Rocky Linux 10 - Extras                                  15 kB/s | 5.4 kB      00:00
Dependencies resolved.
=====
Package            Architecture Version      Repository      Size
=====
Installing:
postfix            x86_64      2:3.8.5-8.el10 appstream        1.5 M
Installing dependencies:
postfix-lmdb       x86_64      2:3.8.5-8.el10 appstream         25 k
=====
Transaction Summary
=====
Install 2 Packages
```

Рис. 11: Установка Postfix

```
[root@client.nsandryushin.net ~]# dnf -y install s-nail
Last metadata expiration check: 0:01:14 ago on Sat 25 Oct 2025 02:41:02 PM UTC.
Dependencies resolved.
=====
Package                Architecture      Version           Repository        Size
=====
Installing:
s-nail                 x86_64            14.9.24-12.el10   appstream         633 k
=====
Transaction Summary
=====
Install 1 Package

Total download size: 633 k
Installed size: 1.2 M
Downloading Packages:
s-nail-14.9.24-12.el10.x86_64.rpm                    1.7 MB/s | 633 kB    00:00
-----
Total                                                  780 kB/s | 633 kB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Running scriptlet: s-nail-14.9.24-12.el10.x86_64 1/1
  Installing      : s-nail-14.9.24-12.el10.x86_64 1/1
  Running scriptlet: s-nail-14.9.24-12.el10.x86_64 1/1
```

Рис. 12: установка s-nail

Настройка протоколов на клиенте

```
[root@client.nsandryushin.net ~]# postfix inet_protocols
postfix/postfix-script: fatal: unknown command: 'inet_protocols'. Usage: postfix start (or
  stop, reload, abort, flush, check, status, set-permissions, upgrade-configuration, logrot
  ate)
[root@client.nsandryushin.net ~]# postconf inet_protocols
inet_protocols = all
[root@client.nsandryushin.net ~]# postconf -e 'inet_protocols = ipv4'
[root@client.nsandryushin.net ~]# systemctl enable postfix
Created symlink '/etc/systemd/system/multi-user.target.wants/postfix.service' → '/usr/lib/
systemd/system/postfix.service'.
[root@client.nsandryushin.net ~]# systemctl start postfix
```

Рис. 13: Настройка протоколов на клиенте

```
[nsandryushin@client.nsandryushin.net ~]$ echo . | mail -s test1 nsandryushin@server.nsandryushin.net
```

Рис. 14: Отправка письма

```
[nsandryushin@server.nsandryushin.net ~]$ tail -f /var/log/maillog
tail: cannot open '/var/log/maillog' for reading: Permission denied
tail: no files remaining
[nsandryushin@server.nsandryushin.net ~]$ sudo tail -f /var/log/maillog
[sudo] password for nsandryushin:
Oct 25 14:32:06 server postfix/postfix-script[10881]: refreshing the Postfix mail system
Oct 25 14:32:07 server postfix/master[10615]: reload -- version 3.8.5, configuration /etc/postfix
Oct 25 14:32:07 server postfix/master[10615]: warning: ignoring inet_protocols parameter value change
Oct 25 14:32:07 server postfix/master[10615]: warning: old value: "all", new value: "ipv4"
Oct 25 14:32:07 server postfix/master[10615]: warning: to change inet_protocols, stop and start Postfix
Oct 25 14:32:48 server postfix/pickup[10885]: 3AAC220444A4: uid=1001 from=<nsandryushin>
Oct 25 14:32:48 server postfix/cleanup[10930]: 3AAC220444A4: message-id=<20251025143248.3AAC220444A4@server.nsandryushin.net>
Oct 25 14:32:48 server postfix/qmgr[10886]: 3AAC220444A4: from=<nsandryushin@nsandryushin.net>, size=358, nrcpt=1 (queue active)
Oct 25 14:32:48 server postfix/local[10933]: 3AAC220444A4: to=<nsandryushin@server.nsandryushin.net>, relay=local, delay=0.12, delays=0.08/0.03/0/0.01, dsn=2.0.0, status=sent (delivered to mailbox)
Oct 25 14:32:48 server postfix/qmgr[10886]: 3AAC220444A4: removed
```

Рис. 15: Логи почты

Включение получения писем из локальной сети

```
[root@server.nsandryushin.net ~]# postconf inet_interfaces
inet_interfaces = localhost
[root@server.nsandryushin.net ~]# postconf mynetworks
mynetworks = 127.0.0.1/32
[root@server.nsandryushin.net ~]# postconf -e 'inet_interfaces = all'
[root@server.nsandryushin.net ~]# postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
[root@server.nsandryushin.net ~]# postfix check
[root@server.nsandryushin.net ~]# systemctl reload postfix
[root@server.nsandryushin.net ~]# systemctl stop postfix
[root@server.nsandryushin.net ~]# systemctl start postfix
```

Рис. 16: Включение получения писем из локальной сети

```
[nsandryushin@client.nsandryushin.net ~]$ echo . | mail -s test1 nsandryushin@server.nsandryushin.net
```

Рис. 17: Повторная отправка письма

Успешная отправка письма согласно логам

```
Oct 25 14:51:31 server postfix/postfix-script[11363]: refreshing the Postfix mail system
Oct 25 14:51:31 server postfix/master[10615]: reload -- version 3.8.5, configuration /etc/postfix
Oct 25 14:51:31 server postfix/master[10615]: warning: ignoring inet_protocols parameter value change
Oct 25 14:51:31 server postfix/master[10615]: warning: old value: "all", new value: "ipv4"
Oct 25 14:51:31 server postfix/master[10615]: warning: to change inet_protocols, stop and start Postfix
Oct 25 14:51:31 server postfix/master[10615]: warning: service smtp: ignoring inet_interfaces change
Oct 25 14:51:31 server postfix/master[10615]: warning: to change inet_interfaces, stop and start Postfix
Oct 25 14:51:36 server postfix/postfix-script[11380]: stopping the Postfix mail system
Oct 25 14:51:36 server postfix/master[10615]: terminating on signal 15
Oct 25 14:51:41 server postfix/postfix-script[11463]: starting the Postfix mail system
Oct 25 14:51:41 server postfix/master[11465]: daemon started -- version 3.8.5, configuration /etc/postfix
Oct 25 14:51:46 server postfix/smtpd[11470]: connect from client.nsandryushin.net[192.168.1.30]
Oct 25 14:51:46 server postfix/smtpd[11470]: EBE862044889: client=client.nsandryushin.net[192.168.1.30]
Oct 25 14:51:46 server postfix/cleanup[11474]: EBE862044889: message-id=<20251025145146.8F28261B7580@client.nsandryushin.net>
Oct 25 14:51:46 server postfix/qmgr[11467]: EBE862044889: from=<nsandryushin@client.nsandryushin.net>, size=597, nrcpt=1 (queue active)
Oct 25 14:51:46 server postfix/smtpd[11470]: disconnect from client.nsandryushin.net[192.168.1.30] ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Oct 25 14:51:47 server postfix/local[11475]: EBE862044889: to=<nsandryushin@server.nsandryushin.net>, relay=local, delay=0.06, delays=0.03/0.02/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Oct 25 14:51:47 server postfix/qmgr[11467]: EBE862044889: removed
```

Рис. 18: Успешная отправка письма согласно логам

```
[nsandryushin@client.nsandryushin.net ~]$ echo . | mail -s test2 nsandryushin@nsandryushin.net
```

Рис. 19: Отправка письма пользователю

Неуспешная доставка письма

```
Oct 25 14:53:12 server postfix/cleanup[11474]: 7482D204488A: message-id=<20251025145312.7482D204488A@server.nsandryushin.net>  
Oct 25 14:53:12 server postfix/qmgr[11467]: 7482D204488A: from=<>, size=2659, nrcpt=1 (queue active)  
Oct 25 14:53:12 server postfix/bounce[11498]: 5E0A42044889: sender non-delivery notification: 7482D204488A  
Oct 25 14:53:12 server postfix/qmgr[11467]: 5E0A42044889: removed  
Oct 25 14:53:12 server postfix/smtp[11497]: connect to client.nsandryushin.net[192.168.1.30]:25: No route to host  
Oct 25 14:53:12 server postfix/smtp[11497]: 7482D204488A: to=<nsandryushin@client.nsandryushin.net>, relay=none, delay=0.01, delays=0/0/0/0, dsn=4.4.1, status=deferred (connect to client.nsandryushin.net[192.168.1.30]:25: No route to host)
```

Рис. 20: Неуспешная доставка письма


```
[root@server.nsandryushin.net ~]# postqueue -p
-Queue ID- --Size-- ----Arrival Time---- -Sender/Recipient-----
7482D204488A      2659 Sat Oct 25 14:53:12  MAILER-DAEMON
      (connect to client.nsandryushin.net[192.168.1.30]:25: No route to host)
                                nsandryushin@client.nsandryushin.net

-- 2 Kbytes in 1 Request.
```

Рис. 21: postqueue -p

Файл прямой зоны

```
GNU nano 8.1 /var/named/master/fz/nsandryushin.net
$TTL 86400 ; 1 day
@      IN SOA  @ server.nsandryushin.net. (
                                2025092014 ; serial
                                86400      ; refresh (1 day)
                                3600       ; retry (1 hour)
                                604800     ; expire (1 week)
                                10800      ; minimum (3 hours)
                                )
      NS   @
      A    192.168.1.1
      MX 10 mail.nsandryushin.net.
      AAAA ::1
$ORIGIN nsandryushin.net.
$TTL 1200 ; 20 minutes
client      A      192.168.1.30
            DHCID  ( AAEBWH0317DS5G/2dhsaE/YY19eqf3aqET2vQfn6RTpz
                    JX4= ) ; 1 1 32
$TTL 86400 ; 1 day
dhcp        A      192.168.1.1
ns          A      192.168.1.1
server      A      192.168.1.1
www         A      192.168.1.1
mail        A      192.168.1.1
```

Рис. 22: Файл прямой зоны

Файл обратной зоны

```
GNU nano 8.1 /var/named/master/rz/192.168.1
$TTL 86400      ; 1 day
@               IN SOA  @ server.nsandryushin.net. (
                                2025092010 ; serial
                                86400      ; refresh (1 day)
                                3600       ; retry (1 hour)
                                604800     ; expire (1 week)
                                10800      ; minimum (3 hours)
                                )
                NS      @
                A       192.168.1.1
                PTR     server.nsandryushin.net.
                MX 10   mail.nsandryushin.net.
                AAAA    ::1
$ORIGIN 1.168.192.in-addr.arpa.
1                PTR     server.nsandryushin.net.
1                PTR     ns.nsandryushin.net.
1                PTR     dhcp.nsandryushin.net.
1                PTR     www.nsandryushin.net.
1                PTR     mail.nsandryushin.net.
$TTL 1200       ; 20 minutes
30              PTR     client.nsandryushin.net.
                DHCID   ( AAEBWH0317DS5G/2dhsaE/YY19eqf3aqET2vQfn6RTpz
                        JX4= ) ; 1 1 32
```

Рис. 23: Файл обратной зоны

Обновление списков назначения почты

```
[root@server.nsandryushin.net ~]# postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain'
[root@server.nsandryushin.net ~]# postfix check
[root@server.nsandryushin.net ~]# systemctl reload postfix
[root@server.nsandryushin.net ~]# restorecon -vR /etc
[root@server.nsandryushin.net ~]# restorecon -vR /var/named
[root@server.nsandryushin.net ~]# systemctl restart named
[root@server.nsandryushin.net ~]# postqueue -f
[root@server.nsandryushin.net ~]# nano /var/named/master/rz/192.168.1
[root@server.nsandryushin.net ~]# nano /var/named/master/rz/192.168.1
[root@server.nsandryushin.net ~]# nano /var/named/master/fz/nsandryushin.net
[root@server.nsandryushin.net ~]# postqueue -f
```

Рис. 24: Обновление списков назначения почты

Лог об успешно доставленном письме

```
Oct 25 14:54:40 server postfix/cleanup[11474]: B456B204488B: message-id=<20251025144724.9764F61B81E4@client.nsandryushin.net>  
Oct 25 14:54:40 server postfix/qmgr[11467]: B456B204488B: from=<nsandryushin@client.nsandryushin.net>, size=597, nrcpt=1 (queue active)  
Oct 25 14:54:40 server postfix/smtpd[11470]: disconnect from client.nsandryushin.net[192.168.1.30] ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7  
Oct 25 14:54:40 server postfix/local[11517]: B456B204488B: to=<nsandryushin@server.nsandryushin.net>, relay=local, delay=0.02, delays=0.01/0/0/0, dsn=2.0.0, status=sent (delivered to mailbox)  
Oct 25 14:54:40 server postfix/qmgr[11467]: B456B204488B: removed
```

Рис. 25: Лог об успешно доставленном письме

Сохранение конфигурации

```
[root@server.nsandryushin.net ~]# cd /vagrant/provision/server/dns/var/named
[root@server.nsandryushin.net named]# cp -R /var/named/* /vagrant/provision/server/dns/var/named
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run-20250927'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/nsandryushin.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.ca'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.empty'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.localhost'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.loopback'? y
[root@server.nsandryushin.net named]# y
bash: y: command not found...
[root@server.nsandryushin.net named]# cd /vagrant/provision/server
[root@server.nsandryushin.net server]# touch mail.sh
[root@server.nsandryushin.net server]# chmod +x mail.sh
[root@server.nsandryushin.net server]# nano mail.sh
```

Рис. 26: Сохранение конфигурации

```
GNU nano 8.1 mail.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Copy configuration files"
#cp -R /vagrant/provision/server/mail/etc/* /etc
echo "Configure firewall"
firewall-cmd --add-service=smtp --permanent
firewall-cmd --reload
restorecon -vR /etc
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
echo "Configure postfix"
postconf -e 'mydomain = user.net'
postconf -e 'myorigin = $mydomain'
postconf -e 'inet_protocols = ipv4'
postconf -e 'inet_interfaces = all'
postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain'
postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
postfix set-permissions
restorecon -vR /etc
systemctl stop postfix
systemctl start postfix
```

Рис. 27: mail.sh

Создание скрипта mail.sh для клиента

```
[root@client.nsandryushin.net ~]# cd /vagrant/provision/client  
[root@client.nsandryushin.net client]# touch mail.sh  
[root@client.nsandryushin.net client]# chmod +x mail.sh  
[root@client.nsandryushin.net client]# nano mail.sh  
[root@client.nsandryushin.net client]# █
```

Рис. 28: Создание скрипта mail.sh для клиента


```
GNU nano 8.1 mail.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Configure postfix"
postconf -e 'inet_protocols = ipv4'
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
```

Рис. 29: mail.sh для клиента

```
94         type: "shell",
95         preserve_order: true,
96         path: "provision/server/mysql.sh"
97     server.vm.provision "server firewall",
98         type: "shell",
99         preserve_order: true,
100         path: "provision/server/firewall.sh"
101     server.vm.provision "server mail",
102         type: "shell",
103         preserve_order: true,
104         path: "provision/server/mail.sh"
105
106 end
107
108 ## Client configuration
109 config.vm.define "client", autostart: false do |client|
110     client.vm.box = "rockylinux10"
111     client.vm.hostname = "client"
112
113     client.vm.boot_timeout = 1440
114
115     client.ash.insert_key = false
116     client.ash.username = "vagrant"
117     client.ash.password = "vagrant"
118
119     client.vm.network :private_network,
120         type: "dhcp",
121         virtualbox____intnet: true
122
123     client.vm.provider :virtualbox do |virtualbox|
124         virtualbox.customize ["modifyvm", :id, "--vrdm", "on"]
125         virtualbox.customize ["modifyvm", :id, "--vrdmport", "3392"]
126     end
127
128     client.vm.provision "client dummy",
129         type: "shell",
130         preserve_order: true,
131         path: "provision/client/01-dummy.sh"
132
133     client.vm.provision "client routing",
134         type: "shell",
135         preserve_order: true,
136         run: "always",
137         path: "provision/client/01-routing.sh"
138
139     client.vm.provision "client mail",
140         type: "shell",
141         preserve_order: true,
142         path: "provision/client/mail.sh"
143 end
144
```

Рис. 30: vagrantfile

В результате выполнения лабораторной работы были получены навыки настройки smtp сервера